Vikas Pathade

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Objective

Aspiring AI & Data Science student with hands-on experience in LLM orchestration, multimodal deep learning, and developer analytics. Seeking opportunities to engineer production-ready AI systems and extract actionable insights from complex data.

Education

B.Tech in Artificial Intelligence & Data Science — 2023 – 2027
CSMSS Chh. Shahu College of Engineering — *CGPA: 7.98/10*

Higher Secondary Certificate (HSC) — 2022
Maharashtra State Board — *81.67%*

Secondary School Certificate (SSC) — 2020
Maharashtra State Board — *92%*

Technical Skills

Languages: Python, SQL, C
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, NLTK, Flask, FastAPI, LangChain
Core Concepts: Machine Learning, Deep Learning, NLP, Computer Vision, Transformer Models (BERT, WavLM), Co-attention, LLM APIs, Multi-Agent Systems
Tools & Platforms: Git, GitHub, Jupyter, VS Code, Render, Streamlit, Vercel, Librosa, Whisper

Projects

Multi-Agent Research System | GitHub | Live Demo — June 2026 – Present

- Orchestrated **4 collaborative agents** (Search, Reader, Writer Chain, Critic Chain) using LangChain & Google Gemini 2.5 Flash – achieves **10-12s pipeline** with self-critique scores **3/10 to 8/10**.
- Implemented **self-improvement loop** (Critic → Revision → Critic, 2 cycles max) and **parallel async scraping** (2x faster); integrated **quota tracking** (18/20 requests remaining) with structured Pydantic output.
- Each search covers **5 URLs** (2 scraped); deployed on Streamlit Cloud with session persistence.
- **Tech:** Python, LangChain, Gemini API, Streamlit, Tavily, AsyncIO, Pydantic — **Deployed:** Streamlit

Speech-based Alzheimer's Disease Detection | GitHub — Feb 2026 – May 2026

- Fused WavLM (512-d), BERT (768-d) & 4 clinical features via co-attention, achieving **89.3% validation accuracy** and **83.4% 5-fold CV** on DementiaBank Pitt Corpus.
- Designed co-attention network with 2 transformer blocks (5.7M parameters) and optimal threshold 0.43 for uncertainty-aware prediction.
- Built Flask prototype with Whisper transcription to extract disfluency biomarkers (pause count, filler rate, pitch variability).
- **Tech:** Python, PyTorch, BERT, WavLM, Co-attention, Librosa, Flask, Whisper

DevInsightAI | GitHub | Live Demo — Dec 2025 – Feb 2026

- Analyzed GitHub profiles via API (commits, PRs, issues) to build a productivity scoring system; added **profile comparison feature** for side-by-side benchmarking.
- Integrated LLM APIs (OpenAI) to generate natural language summaries for each developer profile.
- Deployed on Vercel with FastAPI backend; serves real-time analytics.
- **Tech:** Python, FastAPI, GitHub API, OpenAI API, HTML/CSS/JS — **Deployed:** Vercel

Achievements & Certifications

500+ LeetCode problems | Machine Learning (IBM) | Deep Learning (Infosys) | NLP (Infosys) | Computer Vision (Infosys) | Prompt Engineering (Infosys)